

Econometrics I

Review: Hypothesis Testing for Population Means

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Estimation of the Population Mean

This set of slides is intended as a review of hypothesis testing and constructing confidence intervals. You should have seen this in ECON 211/MATH 321-322.

- Consider population described by the random variables Y and ϵ and suppose that we have access to a random sample of size n of such population. The observations satisfy:

$$y_i = \mu_Y + \epsilon_i, \quad i = 1, \dots, n$$

where the collection of random variables (y_i, ϵ_i) $i = 1, \dots, n$ are independent and identically distributed, since they come from a random sample from the same parent population.

- Furthermore, suppose that the following assumptions are true for ϵ :
 - 1 $E(\epsilon) = 0$
 - 2 $Var(\epsilon) = \sigma^2$
 - 3 $\epsilon \sim N(0, \sigma^2)$
- Then $E(Y) = \mu_Y$ and $\sigma_Y^2 = Var(Y) = \sigma^2$, and $Y \sim N(\mu_Y, \sigma^2)$.

As you can see, the previous is a **simplified** linear regression model, which satisfies assumptions 1 to 4 from lecture 3. The simplification lies in the fact that Y is assumed to be independent of the explanatory variable X .

Properties of $\bar{Y}$

- The sample mean $\bar{Y}$ is an unbiased estimator of μ_Y .
- The sample mean $\bar{Y}$ is the best linear unbiased estimator where “best” stands for more efficient here.
- The sample mean $\bar{Y}$ is also the least squares estimator of μ_Y .

Hypothesis Tests About μ_Y

The objective is to make a provisional decision based on the evidence at hand whether a *null* hypothesis is true, or instead that some *alternative* hypothesis is true. That is, test

- ① $H_0 : E[Y] = \mu_{Y,0}$ versus $H_I : E[Y] > \mu_{Y,0}$ (1-sided $>$).
- ② $H_0 : E[Y] = \mu_{Y,0}$ versus $H_I : E[Y] < \mu_{Y,0}$ (1-sided $<$).
- ③ $H_0 : E[Y] = \mu_{Y,0}$ versus $H_I : E[Y] \neq \mu_{Y,0}$ (2-sided).

Here, $\mu_{Y,0}$ denotes the hypothesised value of the population mean (i.e., the value of the population mean “under” the null hypothesis).

The *p-value*

The p-value is the probability of drawing a statistic at least as adverse to the null hypothesis as the value actually computed with your data, assuming that the null hypothesis is true. One often “rejects the null hypothesis” when the p-value is less than the significance level α .

The *significance level* α

The significance level of a test is a pre-specified probability of incorrectly rejecting the null, when the null is true.

Hypothesis Tests Concerning the Population Mean

Example

- Suppose we are analyzing the height of the population of students at Nazarbayev University, and that we believe the average height is 1.75 mts. This is our null hypothesis.
- We draw a random sample and we find that the average height in the sample is 1.8 mts.
- The p-value is the probability of observing a sample mean $\bar{Y} = 1.8$ mts. when the true population average is 1.75 mts.
- If the p-value is small (for example 0.05), then it is very unlikely that a random sample would lead to an estimate of 1.8 mts when the true population value is 1.75 mts.

Calculation of the *p-value*

Let $\bar{Y}^{act}$ be the value of the sample average “actually” computed with the sample data and let Pr_{H_0} denote the probability computed under the null hypothesis (i.e., computed assuming that $E[Y] = \mu_{Y,0}$). Then, the p-value is

$$\text{p-value} = Pr_{H_0} \left(|\bar{Y} - \mu_{Y,0}| > |\bar{Y}^{act} - \mu_{Y,0}| \right) \quad \text{two-sided} \quad (1)$$

$$\text{p-value} = Pr_{H_0} \left((\bar{Y} - \mu_{Y,0}) > (\bar{Y}^{act} - \mu_{Y,0}) \right) \quad \text{one-sided} \quad (2)$$

Hypothesis Tests Concerning The Population Mean

Computing the p-value for n Large and σ_Y Known

$$\begin{aligned}\text{p-value} &= Pr_{H_0} \left(|\bar{Y} - \mu_{Y,0}| > |\bar{Y}^{act} - \mu_{Y,0}| \right), \\ &= Pr_{H_0} \left(\left| \frac{\bar{Y} - \mu_{Y,0}}{\sigma_{\bar{Y}}} \right| > \left| \frac{\bar{Y}^{act} - \mu_{Y,0}}{\sigma_{\bar{Y}}} \right| \right), \\ &= Pr_{H_0} \left(Z_0 > \left| \frac{\bar{Y}^{act} - \mu_{Y,0}}{\sigma_{\bar{Y}}} \right| \right), \\ &= 2\Phi \left(- \left| \frac{\bar{Y}^{act} - \mu_{Y,0}}{\sigma_{\bar{Y}}} \right| \right).\end{aligned}\tag{3}$$

where $\sigma_{\bar{Y}} = \sigma_Y / \sqrt{n}$, and where $\Phi(z) = \Pr(Z \leq z)$, i.e., the probability that a standard normal random variable Z is less than or equal to z .

Hypothesis Tests Concerning The Population Mean

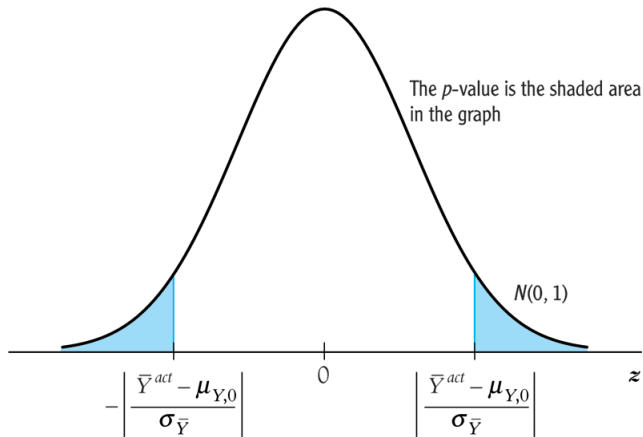


Figure: p -value for known variance.

Hypothesis Tests Concerning the Population Mean

Example

- Suppose we know that the standard deviation in the distribution of height among NU students is 0.3 mts (i.e., $\sigma_Y = 0.3$).
- In a sample of 100 students the computed sample average height is $\bar{Y} = 1.8$ mts.
- Compute the p-value for the tests
 - ① $H_0 : E[Y] = \mu_{Y,0} = 1.75$ versus $H_I : E[Y] > 1.75$ (1-sided $>$).
 - ② $H_0 : E[Y] = \mu_{Y,0} = 1.75$ versus $H_I : E[Y] \neq 1.75$ (2-sided).

Remarks

- However, to proceed in the way we suggested above, we still need to know the population variance.
- In general the population variance is unknown and it must be estimated before the p-value can be computed.

Sample Variance and Sample Standard Deviation

The sample variance s_Y^2 is

$$s_Y^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2. \quad (4)$$

The sample standard deviation s_Y is the square root of the sample variance. The sample variance is an unbiased estimator of the population variance for the population model in which $Y = \mu_Y + \epsilon$.

Standard Error of $\bar{Y}$

The standard error of $\bar{Y}$ is an estimator of the standard deviation of $\bar{Y}$ and is denoted by $SE(\bar{Y})$ or $\bar{s}$. When $Y_1, \dots, Y_n$ are i.i.d. we have

$$SE(\bar{Y}) = \hat{s} = \frac{s_Y}{\sqrt{n}}. \quad (5)$$

Computing the p-value for n Large and σ_Y Unknown

Since the sample variance is an unbiased estimator of the population variance, the p-value can be computed by using $SE(\bar{Y})$. Thus, when σ_Y is unknown we can use the formula

$$\text{p-value} = 2\Phi\left(-\left|\frac{\bar{Y}^{act} - \mu_{Y,0}}{SE(\bar{Y})}\right|\right), \quad (6)$$

where $SE(\bar{Y}) = \hat{s} = \frac{s_Y}{\sqrt{n}}$.

The t-statistic

The standardized sample average $(\bar{Y} - \mu_{Y,0}) / SE(\bar{Y})$ plays an important role in testing statistical hypotheses and it is called the *t-statistic* or *t-ratio*,

$$t = \frac{\bar{Y} - \mu_{Y,0}}{SE(\bar{Y})}. \quad (7)$$

We will see later that the *t*-statistic, under the previous assumptions, has a ***t*-Student Distribution**. However, we will now proceed as follows for pedagogical purposes: when n is very large, the sample variance is close to the population variance, and thus the distribution of the *t*-statistic can be **well-approximated** by the standard normal.

Later in the course we will see the formal meaning of **well-approximated**.

The t-statistic and the p-value

We can write the formula for the p-value in terms of the t-statistic. Let t^{act} denote the value of the t-statistic actually computed

$$t^{act} = \frac{\bar{Y}^{act} - \mu_{Y,0}}{SE(\bar{Y})}. \quad (8)$$

Hence, when n is large the p-value can be computed by

$$\text{p-value} = 2\Phi(-|t^{act}|) \quad \text{two-sided}, \quad (9)$$

$$\text{p-value} = 1 - \Phi(t^{act}) \quad \text{one-sided } >, \quad (10)$$

$$\text{p-value} = \Phi(t^{act}) \quad \text{one-sided } <. \quad (11)$$

Hypothesis Tests Concerning the Population Mean

Example

- Suppose the standard deviation in the distribution of height among NU students is unknown (σ_Y unknown).
- In a random sample of 100 students the computed sample average height is $\bar{Y}^{act} = 1.8$ mts and the sample variance is $s^2 = 0.08$.
- Compute the p-value for the test $H_0 : E[Y] = \mu_{Y,0} = 1.75$ versus $H_I : E[Y] > 1.75$.

Hypothesis Tests Concerning the Population Mean

Hypothesis Testing With a Prespecified Significance Level

Type I Error

Reject the null hypothesis when it is true.

Type II Error

Not rejecting the null hypothesis when it is false.

Significance Level

The significance level of the test is the probability of rejecting the null hypothesis when it is true (probability of type I error). Hence, another way to think about the p-value is as the smallest significance level at which you can reject the null hypothesis.

Hypothesis Testing With a Prespecified Significance Level

Critical Value

The critical value of the test is the value of the test statistic for which the test just rejects the null hypothesis at the chosen significance level.

Rejection Region

The rejection region is the set of values of the test statistic for which the null hypothesis is rejected.

Acceptance Region

The acceptance region is the set of values of the test statistic for which the null hypothesis is not rejected.

Hypothesis Testing With a Prespecified Significance Level

Size of the Test

The size of the test is the probability of type I error.

Power of the Test

The power of the test is the probability of rejecting the null hypothesis when the alternative is true.

Hypothesis Testing With a Prespecified Significance Level

How Do We Choose the Significance Level?

- Econometricians typically use a 5% significance level, which means that they are incorrectly rejecting the null hypothesis on average 1 out of 20 times.
- Using a very low significance level makes more difficult to reject the null hypothesis when it is false (i.e., the lower the significance level the lower the power of the test).

Implementing a Two-Sided Test on the Population Mean

- 1 Compute $SE(\bar{Y})$ (standard error of $\bar{Y}$).
- 2 Compute the t-statistic.
- 3 Compute the p-value as

$$\text{p-value} = Pr_{H_0}(Z > |t^{act}|) = 2\Phi(-|t^{act}|).$$

- 4 Reject the null hypothesis at the 5% significance level if the p-value is less than 0.05.
- 5 Equivalently, reject the null hypothesis if $|t^{act}| > 1.96$, where 1.96 is the critical value of the t statistic (i.e., the value that leaves 2.5% of the probability mass to the right).

Implementing a One-Sided ($>$) Test on the Population Mean

- 1 Compute $SE(\bar{Y})$ (standard error of $\bar{Y}$).
- 2 Compute the t-statistic.
- 3 Compute the p-value as

$$\text{p-value} = Pr_{H_0} (Z > t^{act}) = 1 - \Phi(t^{act}).$$

- 4 Reject the null hypothesis at the 5% significance level if the p-value is less than 0.05.
- 5 Equivalently, reject the null hypothesis if $t^{act} > 1.64$, where 1.64 is the critical value of the t statistic (i.e., the value that leaves 5% of the probability mass to the right).

- Even though we cannot learn the true value of μ_Y (the population mean of Y) using only a sample, we can construct an interval containing μ_Y with a certain prespecified probability.
- This interval is called *confidence interval*, and the probability level is called the *confidence level*.
- For example, a 95% confidence interval for μ_Y is an interval that contains the true value of μ_Y in 95% of repeated samples.

Some Confidence Intervals

When sample sizes are large the 90%, 95%, and the 99% confidence intervals are

$$90\% \text{ confidence interval for } \mu_Y : [\bar{Y} \pm 1.64 \times SE(\bar{Y})] . \quad (12)$$

$$95\% \text{ confidence interval for } \mu_Y : [\bar{Y} \pm 1.96 \times SE(\bar{Y})] . \quad (13)$$

$$99\% \text{ confidence interval for } \mu_Y : [\bar{Y} \pm 2.58 \times SE(\bar{Y})] . \quad (14)$$

Remarks: What is Random Here?

- The values of $Y_1, \dots, Y_n$ and thus any function of them – including the confidence interval which will differ from one sample to the next.
- The population parameter μ_Y is not random; we just don't know it.

The t-Statistic Testing the Mean

Let Y_i for $i = 1, \dots, n$ be an i.i.d. sample. As we have stated before, the t-statistic for testing $H_0 : E(Y) = \mu_{Y,0}$ is

$$t = \frac{\bar{Y} - \mu_{Y,0}}{\sqrt{s_Y^2/n}} = \frac{\bar{Y} - \mu_{Y,0}}{SE(\bar{Y})}, \quad (15)$$

where $s_Y^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2$ is the sample variance.

The t-Statistic Testing the Mean

- The exact distribution of the t-statistic testing the mean of a single population is the *Student t* distribution with $n - 1$ degrees of freedom.
- When the sample size is small, the standard normal distribution is a poor approximation for the distribution of the t-statistic.

Some Remarks on the Use of the Student t Distribution in Practice

- For testing the mean of Y , the *Student t Distribution* can be used **only** if the underlying population distribution is normal, i.e., if $Y \sim N(\mu_Y, \sigma^2)$.
- If the sample size is moderate (several dozen) or large (hundreds or more) the difference between the t-Distribution and the Standard Normal critical values is negligible.
- Thus, when the sample size is large, the t-statistic distribution can be approximated by a standard normal distribution. This methodology requires much less assumptions than the ones made in the current set of slides and in the assumptions for the simple linear regression model in lecture 3.

Hypothesis Tests Concerning The Population Mean

degrees of freedom ($n - 1$)	5% t -distribution critical value
10	2.23
20	2.09
30	2.04
60	2.00
∞	1.96

Figure: Critical values in the t -Student Distribution

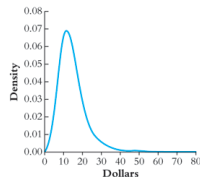
Some Remarks on the Use of the Student t Distribution in Practice

- In sum, the Student-t distribution is only relevant when the sample size is very small; but in that case, for it to be correct, you must be sure that the population distribution of Y is normal.
- In economic data, the normality assumption is rarely credible.

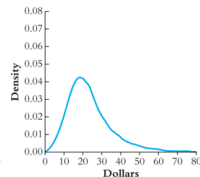
Using the t-Statistic When the Sample Size is Small

FIGURE 2.4 Conditional Distribution of Average Hourly Earnings of U.S. Full-Time Workers in 2008, Given Education Level and Gender

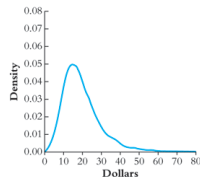
The four distributions of earnings are for women and men, for those with only a high school diploma (a and c) and those whose highest degree is from a four-year college (b and d).



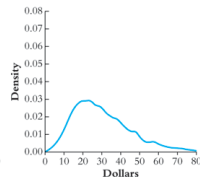
(a) Women with a high school diploma



(b) Women with a college degree



(c) Men with a high school diploma



(d) Men with a college degree

Figure: Economic Data and the Normality Assumption